DAY 14 | POST-READ

Spark and Pipelines: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Medallion layers	Bronze raw, silver clean, gold aggregated. Each rebuildable from the previous.
Production batch jobs	Scheduled, idempotent, partitioned, observable.
Spark architecture	Driver plans; executors run tasks on partitions in parallel.
SparkSession	Entry point: SparkSession.builder.master("local[*]").getOrCreate().
Lazy evaluation	Transformations build a plan; actions (show, count, write) run it.
Transformations	select, filter, withColumn, when/otherwise, dropDuplicates, groupBy().agg().
Joins	inner, left, left_anti, left_semi. Check row counts after joining.
Broadcast join	Send a small table to every executor to avoid shuffling the big one.
Parquet	Columnar, typed, compressed. partitionBy for folder-per-value layout.

Part B: Going beyond today

1. Explicit schemas

```
schema = "order_id INT, customer_id INT, amount DOUBLE, order_date DATE, city STRING"
df = spark.read.csv("orders.csv", header=True, schema=schema)
```

inferSchema reads the data twice and can guess wrong. Production jobs declare the schema.

2. Shuffles

groupBy and joins move data between executors so matching keys meet. This shuffle is the most expensive part of most jobs. Filter early and select only needed columns to shrink it.

3. Seeing the plan

```
daily.explain()   # shows the optimised physical plan
```

4. Window functions

```
from pyspark.sql.window import Window
w = Window.partitionBy("city").orderBy(F.desc("revenue"))
daily.withColumn("rank", F.rank().over(w)).filter("rank = 1")
```

Top item per group, running totals, previous-day comparisons: windows handle them without self-joins.

5. Delta Lake

Delta Lake adds transactions, updates and time travel on top of Parquet files. It is the default table format on Databricks and in many lakehouses.

Part C: Check yourself

- What do the driver and executors each do?
- Why does df.filter(...) return instantly even on a huge file?
- When would you use a left_anti join?
- Why is mode("overwrite") with partitionBy useful for idempotent daily jobs?
- Why is Parquet better than CSV for analytics?

Part D: Practice ideas for tonight

- Add a product dimension and report revenue by category and city.
- Use a window function to find the top city per day.
- Compare the time of the same aggregation in pandas and Spark on 1 million generated rows.

Coming up on Day 15: a full day on FDE professional skills: identity and mindset, and bias towards action, with simulations and role plays.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today